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Flanker Task EEG Functional Data Analysis

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Abstract

This paper describes a functional data analysis (FDA) study of electroencephalography (EEG) data recorded during a Flanker cognitive task. The dataset consists of 62 adults from the cognitive electrophysiology in socioeconomic context dataset, with EEG signals captured at electrode channel FC1 over the time window from $-0.2\,\mathrm{s}$ to $0.8\,\mathrm{s}$ relative to stimulus onset. The raw discrete observations were converted into smooth functional objects using a penalized B-spline basis smoothing. Exploratory data analysis was then performed, including descriptive statistics (mean, standard deviation, covariance), principal component analysis (PCA), and depth-based outlier detection methods. Hypothesis testing revealed no statistically significant differences between groups defined by ADHD status, gender, or socioeconomic status. A function-on-scalar regression model further identified a significant effect of gender on the EEG response shape, with female subjects exhibiting systematically lower amplitudes in the post-stimulus window.

Keywords: Functional Data Analysis, EEG, Flanker task, B-spline smoothing, PCA, Boxplot, Outlier detection, Hypothesis testing, Regression

1. Introduction

Functional Data Analysis is a branch of statistics dealing with data that consists of functions or curves observed over time. The theoretical foundations of Functional Data Analysis are described in the seminal textbook of the same name (see Ramsay and Silverman (2006)). FDA treats each observation as a single functional object rather than a vector of discrete measurements. This perspective is particularly useful in neuroscience, where signals such as EEG are inherently continuous.

The Eriksen Flanker task is a well-established cognitive paradigm used to study attention and response inhibition (see Eriksen and Eriksen (1974)). Participants are presented with a central target stimulus flanked by congruent or incongruent distractors, and EEG recordings capture the brain's electrophysiological response to these stimuli. This project applies FDA techniques to investigate the structure and variability of EEG responses during the Flanker task.

2. Goals and Objectives

The main goal of this project is to demonstrate the application of FDA methods to real-world EEG data from a cognitive electrophysiology experiment. The objectives are as follows:

- To convert raw discrete EEG observations into smooth functional objects using B-spline basis expansion with a roughness penalty.
- To perform exploratory data analysis including computation of sample mean, standard deviation, and covariance functions.
- To identify dominant modes of variation among subjects using Principal Component Analysis (PCA).
- To detect outlier subjects using functional boxplots and depth-based methods.
- To interpret the obtained results in the context of EEG signal analysis.
- To test for statistically significant differences in EEG response curves between groups defined by ADHD status, gender, and socioeconomic status.
- To model the relationship between subject characteristics and EEG response curves using functional regression analysis.

3. Data and Preprocessing

3.1 Dataset

This dataset contains EEG brain recordings from 127 young adults, together with retrospective objective and subjective reports of childhood family socioeconomic status (SES), SES indicators in adulthood, and self-reports of ADHD symptoms. This project applies Functional Data Analysis methods to EEG data from 62 adults recorded during a Flanker task at electrode channel FC1. Data source: Cognitive Electrophysiology in Socioeconomic Context dataset (see Isbell et al. (2025)).

3.2 Smoothing

The first step in any FDA workflow is the conversion of discrete observations into smooth functional objects, in order to reduce measurement noise. A penalized B-spline basis expansion was employed for this purpose, with these specifications:

1. Basis: cubic B-splines (order 4), with 20 basis functions defined over the interval $-0.2\,\mathrm{s}$ to $0.8\,\mathrm{s}$.
2. Penalty: roughness penalty applied to the second derivative.
3. Smoothing parameter: λ selected via Generalized Cross-Validation (GCV).

All analyses were carried out in R, using the `fda`, `fda.usc`, and `refund` packages, with Python equivalents implemented natively using `scikit-fda`, `numpy`, and `scipy`.

The effect of the smoothing procedure is illustrated in Figure 1.

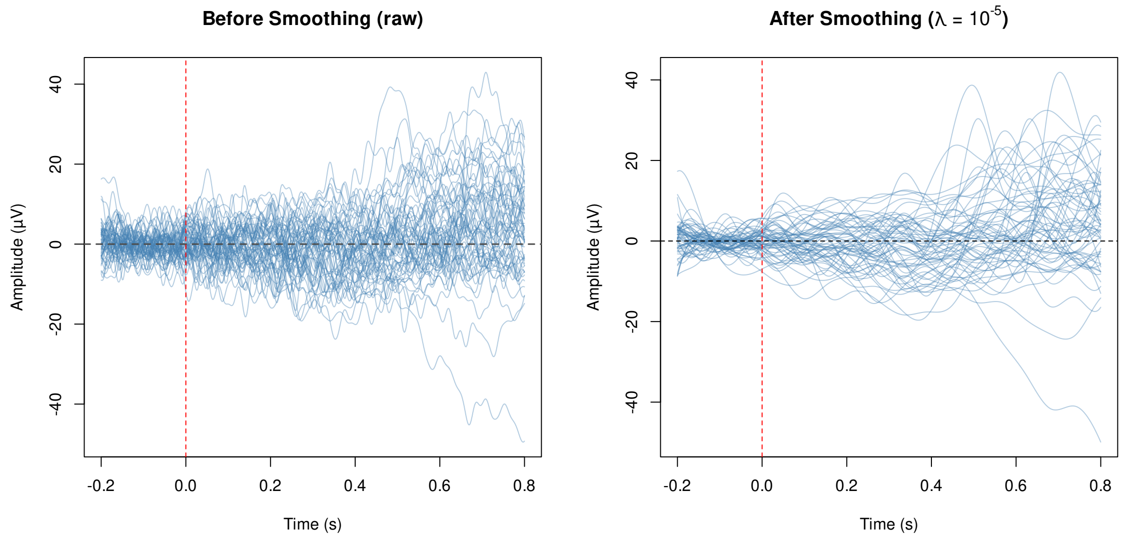


Figure 1: Observed EEG curves before and after B-spline smoothing.

4. Exploratory Data Analysis

4.1 Mean and Standard Deviation

The sample mean and sample standard deviation functions were computed as:

$$\bar{x}(t) = \frac{1}{n}\sum_{i=1}^{n} x_i(t), \qquad \hat{\sigma}(t) = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i(t) - \bar{x}(t))^2} \tag{1}$$

The mean curve exhibits a gradual positive shift following the stimulus onset, while the standard deviation increases from approximately 2 μV during the pre-stimulus baseline to over 10 μV after 0.4 s. This suggests that between-subject variability is substantially larger in the post-stimulus window than during the baseline period. The mean $\pm$ SD band contains the majority of curves, though several subjects clearly deviate beyond it. The sample mean function and pointwise standard deviation are shown in Figure 2.

4.2 Covariance

The sample covariance function quantifies how amplitude values at different time points co-vary across subjects:

$$\hat{c}(s,t) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i(s) - \bar{x}(s)) \cdot (x_i(t) - \bar{x}(t)) \tag{2}$$

The resulting bivariate function was visualized using both perspective and image plots. The diagonal of the covariance surface coincides with the variance function, and the largest values are concentrated in the time region 0.5–0.7 s on both axes. These results suggest that the late post-stimulus response constitutes the dominant source of between-subject

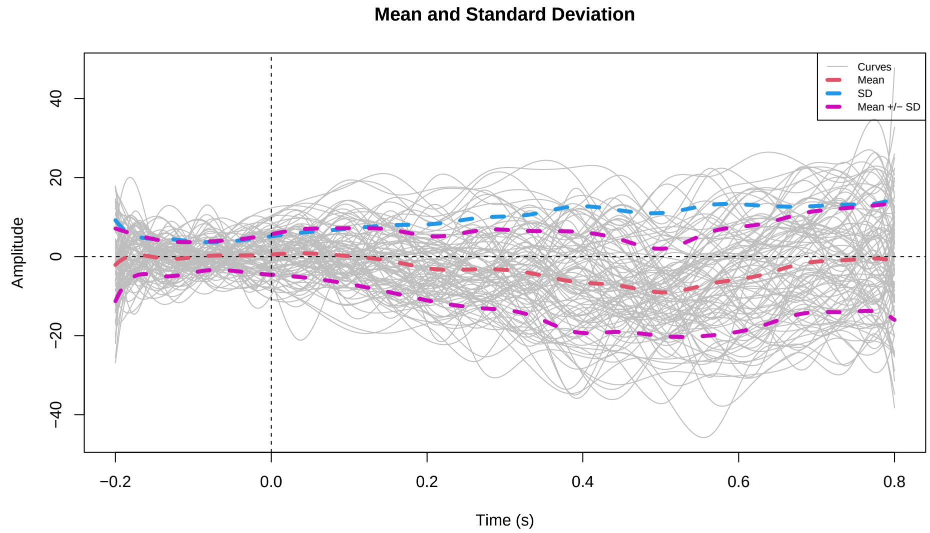
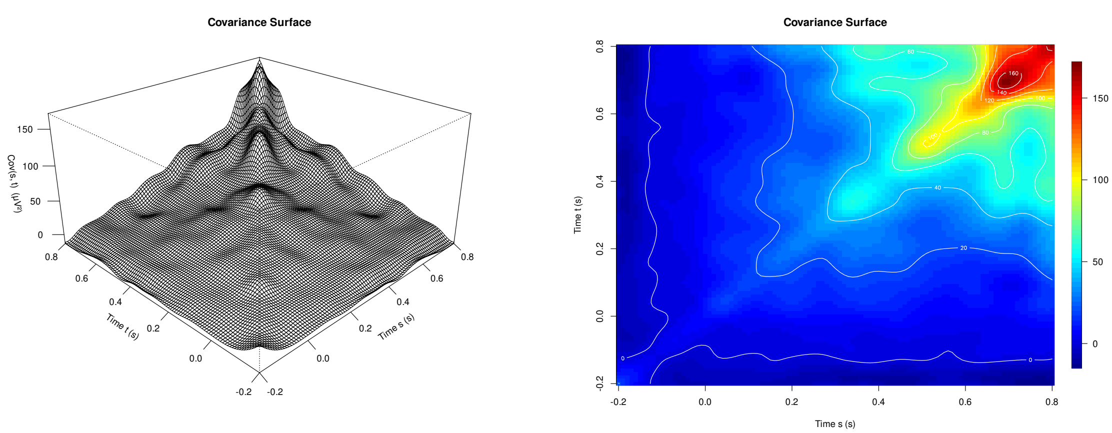


Figure 2: Sample mean function and pointwise standard deviation.

variability, whereas the pre-stimulus baseline shows near-zero covariance, indicating consistent behaviour across subjects during that period. The covariance structure is visualized in Figure 3.



(a) Perspective plot.

(b) Image plot.

Figure 3: Perspective and image visualizations of the sample covariance function.

4.3 Principal Component Analysis

Functional Principal Component Analysis (FPCA) was applied to identify the dominant modes of variation among subjects. The explained variance proportions are:

- PC1: 61.1%

- PC2: 13.6%
- PC3: 11.9%
- PC4: 3.3%

The first three components cumulatively explain 86.6% of the total variability, indicating that the functional data can be effectively summarized by a low-dimensional representation. The first four functional principal component harmonics are displayed in Figure 4.

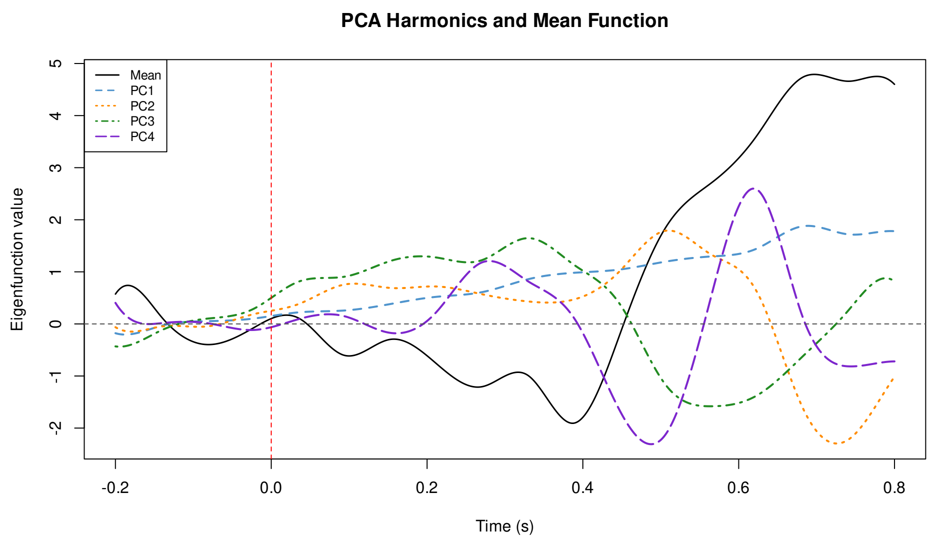


Figure 4: First four functional principal component harmonics.

4.4 Functional Boxplot

Functional boxplots based on Modified Band Depth (MBD) and Band Depth (BD2) were used to identify central and outlying curves. The central 50% region (analogous to the interquartile range in classical boxplots), together with the median curve and the fences beyond which curves are flagged as outliers, were displayed. Both methods identified the same outlier subjects, providing consistent results. The resulting functional boxplot is shown in Figure 5.

4.5 Outliers

Several outlier detection methods were applied, including Highest Density Region (HDR) boxplots and Massive Unsupervised Outlier Detection (MUOD). Four subjects — `sub-003`, `sub-040`, `sub-043`, and `sub-071` — were identified as outliers by the HDR boxplot at coverage level $\alpha = (0.05, 0.50)$. All four exhibit substantial negative deflections in the post-stimulus window: `sub-071` (purple) shows the most extreme deviation, reaching approximately -50 μV after 0.6 s, while `sub-040` (green) and `sub-043` (blue) both descend to about -20 μV in the same time region. `sub-003` (red) shows a less extreme but still atypical negative pattern. These subjects can be classified as magnitude outliers, as their

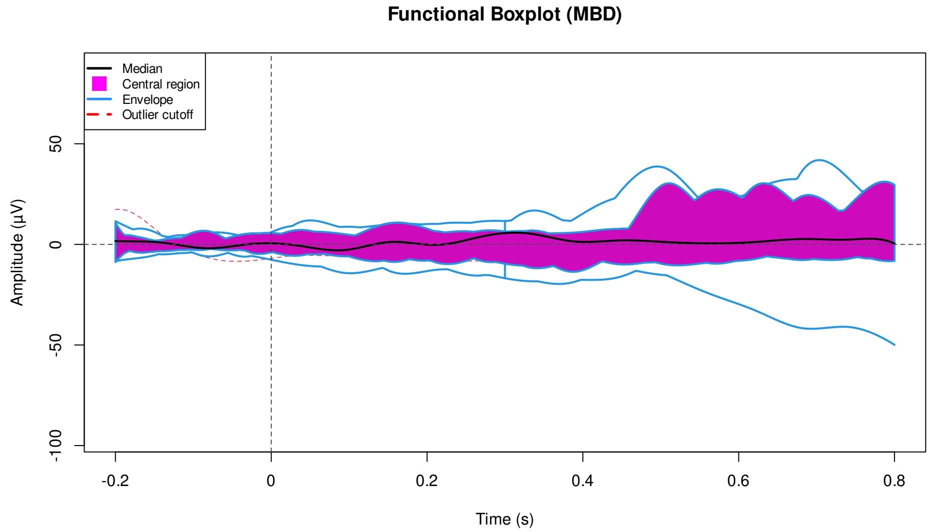


Figure 5: Functional boxplot based on Modified Band Depth (MBD).

waveforms have a similar overall shape to the rest of the sample but are shifted significantly away from the group mean. The four identified outlier subjects were retained in the subsequent hypothesis testing and regression analyses to preserve sample size. The detected magnitude outliers are illustrated in Figure 6.

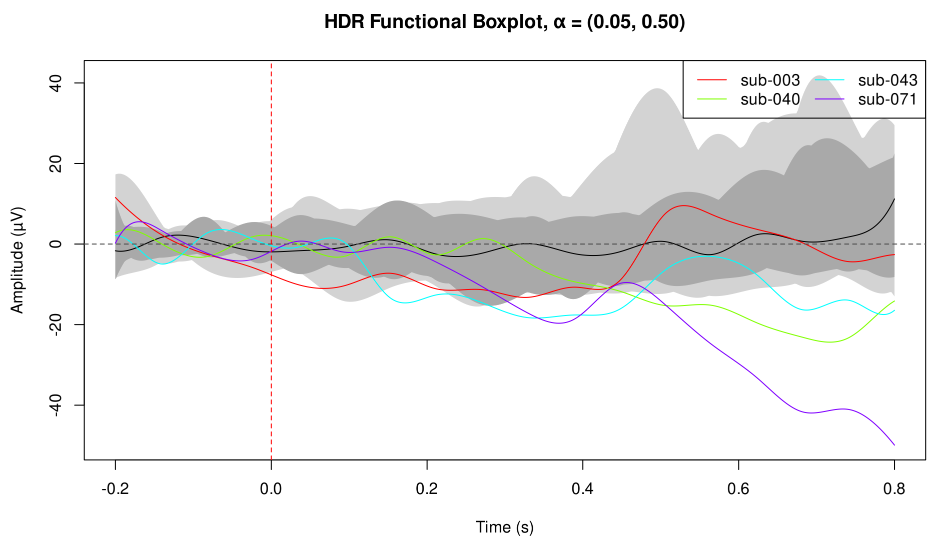


Figure 6: Highest Density Region (HDR) functional boxplot.

5. Hypothesis Testing

The objective of this part of the project is to determine whether the EEG response curves during the Flanker task differ significantly between individuals with ADHD symptoms and those without. The ADHD classification is based on the World Health Organization Adult ADHD Self-Report Scale (ASRS-V1.1), included in the original dataset. Of the 62 subjects analyzed in the previous sections, 12 were classified as having ADHD symptoms, 39 as not having ADHD symptoms, and 11 had missing classifications and were excluded from this analysis. The hypotheses tested are formulated as:

$$H_0 : \mu_{\text{ADHD}}(t) = \mu_{\text{non-ADHD}}(t), \quad \forall\, t \in [-0.2\,\text{s},\, 0.8\,\text{s}] \tag{3}$$

$$H_1 : \mu_{\text{ADHD}}(t) \neq \mu_{\text{non-ADHD}}(t), \quad \text{for some } t \tag{4}$$

5.1 Group Comparison

The 62 smoothed EEG curves were divided into two functional samples according to ADHD classification. The ADHD group consists of 12 subjects, while the non-ADHD group consists of 39 subjects. Visual inspection of the two groups reveals broadly similar patterns of variability around the group mean. Both groups exhibit relatively flat behavior during the pre-stimulus baseline and increased variability in the post-stimulus window. The grouped EEG response curves are presented in Figure 7.

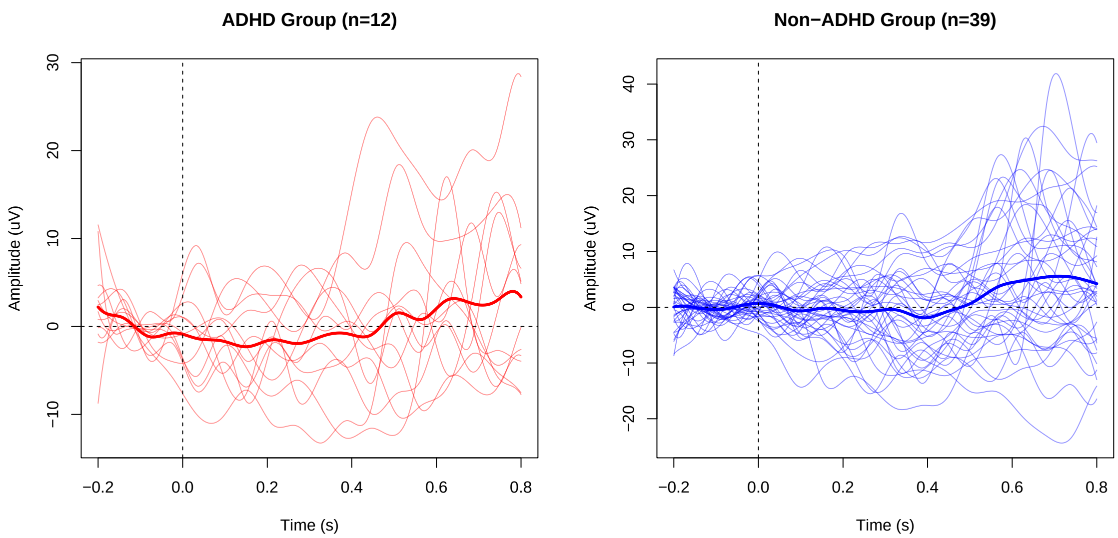


Figure 7: EEG response curves grouped by ADHD classification.

The mean functions of the two groups were computed separately and overlaid for comparison. The non-ADHD mean curve (blue) shows a sustained positive shift after 0.4 s, peaking at approximately $+6\ \mu$V around 0.7 s. The ADHD mean curve (red) follows a similar shape but with a smaller amplitude in the post-stimulus window, reaching approximately $+4\ \mu$V. The group mean functions are compared in Figure 8.

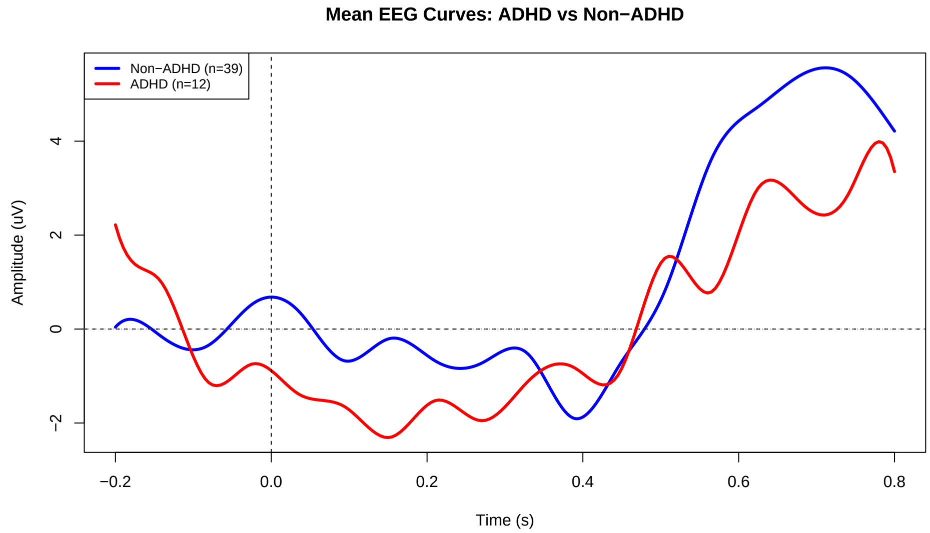


Figure 8: Mean EEG response curves for ADHD and non-ADHD groups.

5.2 Results and Conclusion

Hypothesis tests were applied to assess whether the two group mean functions differ significantly. The results are summarized in Table 1.

Test	**p-value**
Pointwise Z-test ($\max \lvert Z \rvert = 0.66$)	—
L^2-norm test (naive)	1.0000
L^2-norm test (bootstrap)	0.7260
F-type test (naive)	1.0000
F-type test (bootstrap)	0.7140
Permutation test	0.5250

Table 1: Summary of hypothesis test results for ADHD vs. non-ADHD group comparison.

All tests fail to reject the null hypothesis at the significance level $\alpha = 0.05$. Based on this analysis, there is no statistically significant difference between the EEG response curves of individuals with and without ADHD symptoms during the Flanker task at electrode channel FC1.

In addition to the ADHD comparison, hypothesis tests were performed for gender (female vs. male) and socioeconomic status (SES). For the gender comparison, the female group consisted of 37 subjects and the male group of 25 subjects. For the SES comparison, subjects were grouped into low (SES 1–3, $n = 17$), mid (SES 4–6, $n = 29$), and high (SES 7–9, $n = 15$) categories, with one subject excluded due to missing SES data. The results are summarised in Tables 2 and 3.

Test	p-value
Pointwise Z-test ($\max \lvert Z \rvert = 2.50$)	—
L^2-norm test (naive)	0.9931
L^2-norm test (bootstrap)	0.2600
F-type test (naive)	0.9926
F-type test (bootstrap)	0.2680
Permutation test	0.1050

Table 2: Hypothesis test results for female vs. male group comparison.

Test	p-value
Functional one-way ANOVA (bootstrap, 500 replications)	0.4020

Table 3: Hypothesis test results for SES group comparison (low/mid/high).

All tests fail to reject the null hypothesis at $\alpha = 0.05$. There is no statistically significant difference in EEG response curves between female and male subjects, nor between subjects of different socioeconomic backgrounds. Notably, the pointwise Z-test for gender identified 29 time points where $|Z|$ exceeded the critical value of 2.02, suggesting a localised tendency for differences in the post-stimulus window, though this did not reach global significance.

6. Regression

To further investigate the relationship between EEG response curves and subject characteristics, a function-on-scalar regression model was fitted. The regression was performed on the 51 subjects with known ADHD classification, excluding the 11 subjects with missing ADHD status. The model is defined as:

$$Y_i(t) = \beta_0(t) + \beta_1(t) \cdot \mathrm{ADHD}_i + \beta_2(t) \cdot \mathrm{Female}_i + \varepsilon_i(t), \qquad t \in [-0.2\,\mathrm{s},\, 0.8\,\mathrm{s}] \tag{5}$$

The terms used in the regression model are described in Table 4.

6.1 Results and Conclusions

The model summary indicates that the smooth intercept term $\beta_0(t)$ is highly significant ($p = 0.024$), reflecting the structured shape of the average EEG response over time. The ADHD effect $\beta_1(t)$ is not significant ($p = 0.842$), consistent with the findings from the hypothesis testing. The female effect $\beta_2(t)$ is significant ($p = 0.016$), suggesting that gender has a measurable impact on the EEG response shape at channel FC1. The model achieves an adjusted $R^2 = 0.108$, with approximately 11% of the deviance explained. A detailed summary of the fitted regression model is provided in Table 5.

The intercept $\beta_0(t)$ starts at negative values during the pre-stimulus baseline, dips slightly after stimulus onset, and rises to a peak around 0.5 s. The ADHD coefficient

Term	Description	Values
$Y_i(t)$	Smoothed EEG curve of subject i at channel FC1	Continuous
$\beta_0(t)$	Intercept function (baseline mean response)	—
$\beta_1(t)$	Coefficient function for ADHD effect	—
$\beta_2(t)$	Coefficient function for gender effect	—
ADHD_i	Binary indicator of ADHD symptoms	1 = present, 0 = absent
Female_i	Binary indicator of gender	1 = female, 0 = male
$\varepsilon_i(t)$	Residual error function of subject i	—
t	Time relative to stimulus onset	$t \in [-0.2\,\text{s}, 0.8\,\text{s}]$

Table 4: Description of terms in the regression model.

Constant coefficients:				
	Estimate	**Std. Error**	**t value**	**Pr(> $\|t\|$)**
(Intercept)	−0.7201	1.0344	−0.696	0.486
Smooth terms & functional coefficients:				
	edf	**Ref.df**	**F**	**p-value**
Intercept(`yindex`)	10.686	19.000	9.299	0.0237 *
ADHD(`yindex`)	2.001	2.378	0.251	0.8421
Female(`yindex`)	4.498	4.794	2.762	0.0159 *
Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1				
$R^2_{\text{adj}} = 0.108$				Deviance explained = 11.1%

Table 5: Summary of the regression model.

$\beta_1(t)$ fluctuates around zero across the entire time range, visually confirming the non-significant p-value. The female coefficient $\beta_2(t)$ shows a more negative deflection during the post-stimulus window, especially between 0.3 and 0.6 s. This suggests that female subjects exhibit systematically lower amplitudes. The estimated functional regression coefficients are shown in Figure 9.

The observed curves span a wide range from approximately -20 μV to $+40$ μV with substantial individual variability, while the fitted curves are restricted to roughly -6 to $+6$ μV. This visible reduction in range illustrates the low R^2 — the model captures the average behavior driven by ADHD and gender but not other factors not included in the model. A comparison of the observed and fitted EEG curves is presented in Figure 10.

While ADHD status does not significantly affect the EEG response curve, gender shows a statistically significant influence, particularly in the post-stimulus window between 0.3 and 0.6 s. The relatively low explanatory power of the model ($R^2 \approx 0.11$) suggests that additional individual-level factors not included in this analysis contribute substantially to the observed variability in EEG responses.

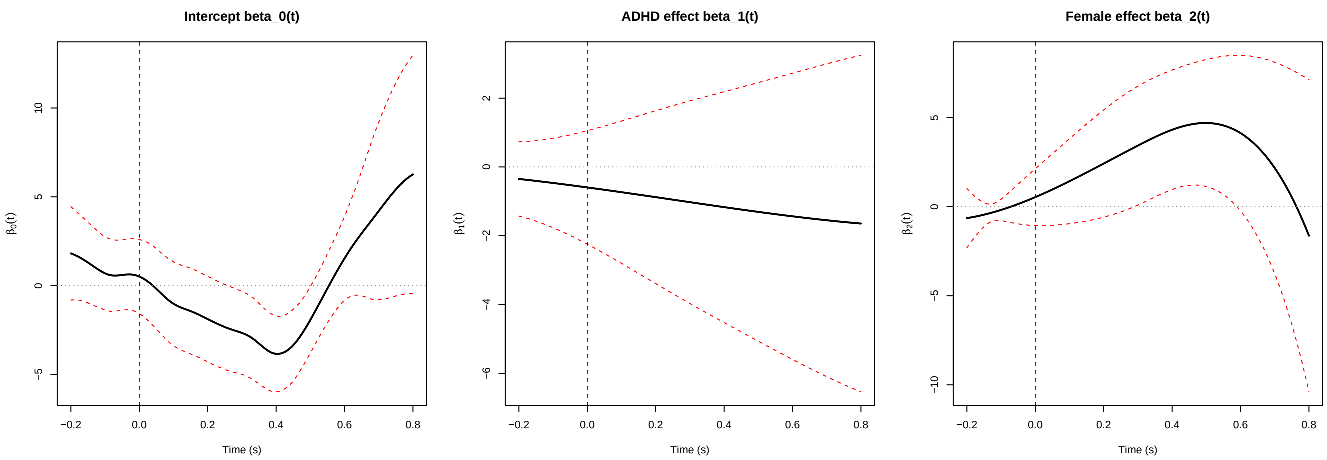


Figure 9: Estimated functional regression coefficients.

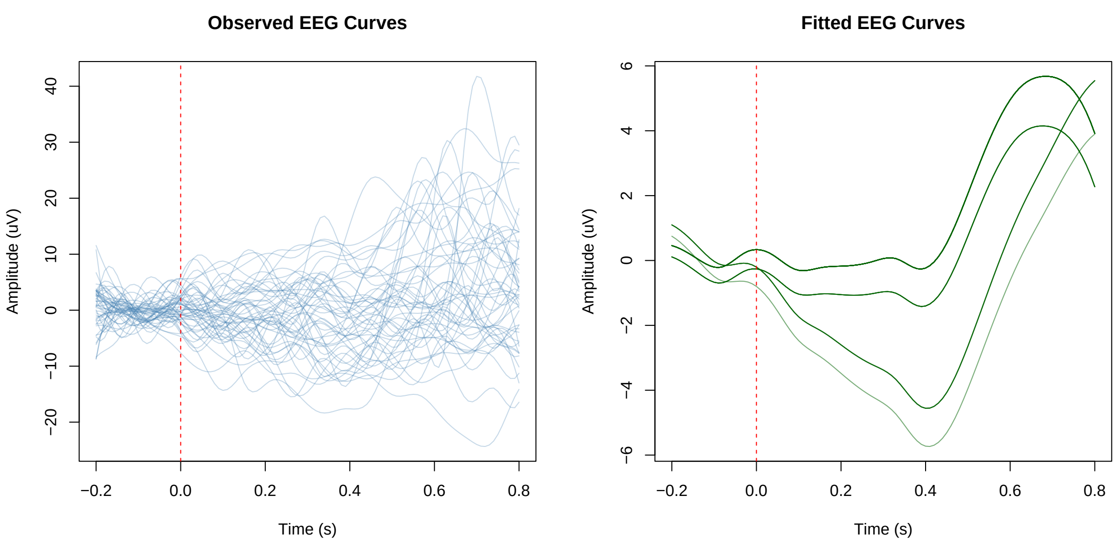


Figure 10: Observed and fitted EEG response curves.

7. Conclusion

This project applied a complete Functional Data Analysis pipeline to EEG data recorded during a Flanker cognitive task at electrode channel FC1, using data from 62 young adults from the publicly available Cognitive Electrophysiology in Socioeconomic Context dataset. In the smoothing step, raw discrete EEG observations were converted into smooth functional objects using a penalized B-spline basis expansion with 20 cubic basis functions. The exploratory data analysis revealed that between-subject variability is concentrated in the post-stimulus window (0.4–0.8 s), with the first three functional principal components explaining 86.6% of the total variance. Outlier detection methods consistently identified a small number of subjects classified as magnitude outliers.

The hypothesis testing procedures investigated whether the mean EEG curves differ between subjects grouped by ADHD status, gender, and socioeconomic status. For the ADHD and gender comparisons, four complementary tests were applied: the pointwise Z-test, L^2-norm test (naive and bootstrap), F-type test (naive and bootstrap), and permutation test. For the three-group SES comparison, a functional one-way ANOVA was used. All tests consistently failed to reject the null hypothesis at $\alpha = 0.05$, suggesting no statistically significant global differences between any of the groups.

The functional regression analysis extended these findings by simultaneously modeling the effects of ADHD status and gender on the EEG curves. ADHD status had no significant effect ($p = 0.842$), confirming the hypothesis testing results. However, gender showed a statistically significant effect ($p = 0.016$), with female subjects exhibiting systematically lower amplitudes in the post-stimulus window between 0.3 and 0.6 s. The relatively low explanatory power of the model ($R^2 \approx 0.11$) suggests that substantial individual variability remains unexplained by the included covariates.

Overall, the project illustrates how FDA methods can be applied to electrophysiological data to extract meaningful insights about functional structure, variability, and covariate effects.

References

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Appendix A. Code Availability

The complete source code used for this project, including the R and Python implementations, data preprocessing scripts, statistical analyses, and LaTeX report source files, is publicly available at:

`https://github.com/00riddle00/Functional-Data-Analysis`

Some FDA procedures were implemented primarily in R due to the greater maturity of FDA-related libraries in the R ecosystem.